



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS
PAPER 2

4004/2
2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Mathematical Instruments
Mathematical Tables/Non programmable Electronic Calculator
Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.
Answer **all questions** in Section A and **any four questions** from Section B.

If you use more than one booklet, fasten them together.

All working must be clearly shown on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

If the degree of accuracy is not specified in the question and if the answer is not exact, the answer should be given correct to three significant figures. Answers in degrees should be given correct to one decimal place.

Mathematical tables and/or Non programmable electronic calculators may be used to evaluate explicit numerical expressions.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

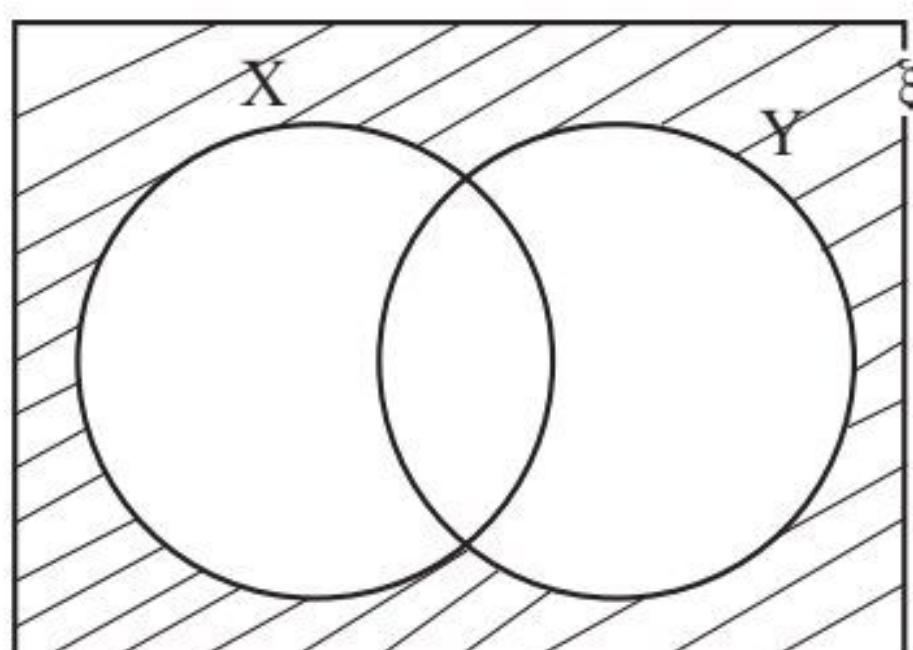
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SECTION A: [52 MARKS]

Answer **all** questions.

- 1 (a) Find the Lowest Common Multiple (LCM) of 24 and 63. [3]
- (b) John and Peter shared \$60,00 in the ratio 7:5 respectively.
Find how much John got more than Peter. [3]
- (c) Simplify $2(a + 3b) - 5(4a - b)$. [3]
- 2 (a) Find the circumference of a circle whose diameter is 3,5 cm.
Take π to be $\frac{22}{7}$. [2]
- (b) For the expressions $12a^3b^2$ and $15a^4b^3$, find the
- (i) Highest Common Factor (HCF),
- (ii) Lowest Common Multiple (LCM). [4]
- (c) (i) Given that $a = 3$, $b = -5$ and $c = 4$.
evaluate $\frac{a-b}{c}$ [2]
- (ii) Solve the equation $3x - 7 = 5x + 9$. [2]
- 3 (a) Calculate the total cost of buying a residential stand on Hire Purchase if an initial deposit of \$2 064 is paid followed by 24 equal monthly instalments of \$164 each. [3]
- (b) The ratio of the radii of two similar cylinders is 2: 7. If the volume of the larger cylinder is $1\,029\text{ cm}^3$, find the volume of the smaller cylinder. [3]
- (c) (i) Jane is x years old. Her brother is 15 years older than her.
Given that the sum of their ages is 45 years, find Jane's age. [2]
- (iii) The Venn diagram shows the universal set ξ with subset X and Y.



Use set notation to describe the shaded region. [2]

- 4 (a) Mr Ndlovu went to South Africa for shopping. He changed \$4 000 ZWD to South African Rands.

Given that the exchange rate was \$1ZWD: R1,44.

Calculate the amount he received in South African Rands. [2]

- (b) Solve the simultaneous equations

$$\frac{1}{3}x = y$$

$$2x - 3y = 1$$

[3]

- (c) Two points P and Q, on level ground are such that Q is to the North-East of P.

(i) State the 3-figure bearing of Q from P. [1]

(ii) From the top of a flag post 12 metres in height, directly on top of point P, the angle of depression of a peg on the ground is 60° . Find the distance of the peg from P. [2]

- (d) Find the number of sides of a regular polygon whose exterior angle is 8° . [3]

- 5 **Answer the whole of this question on a sheet of plain paper.**
Use ruler and compasses only. All constructions should be in a single diagram.

- (a) Construct a quadrilateral ABCD such that $AB = 7$ cm, $BC = 10$ cm, $CD = 8$ cm, $\hat{ABC} = 90^\circ$ and $\hat{BCD} = 60^\circ$. [6]

- (b) Measure and write down the length of AD. [1]

- (c) Construct the

(i) locus of points 4 cm from D. [1]

(ii) locus of points equidistant from AB and BC. [2]

- (d) Mark and label two points M and N such that they are 4 cm from D and equidistant from AB and BC. [2]

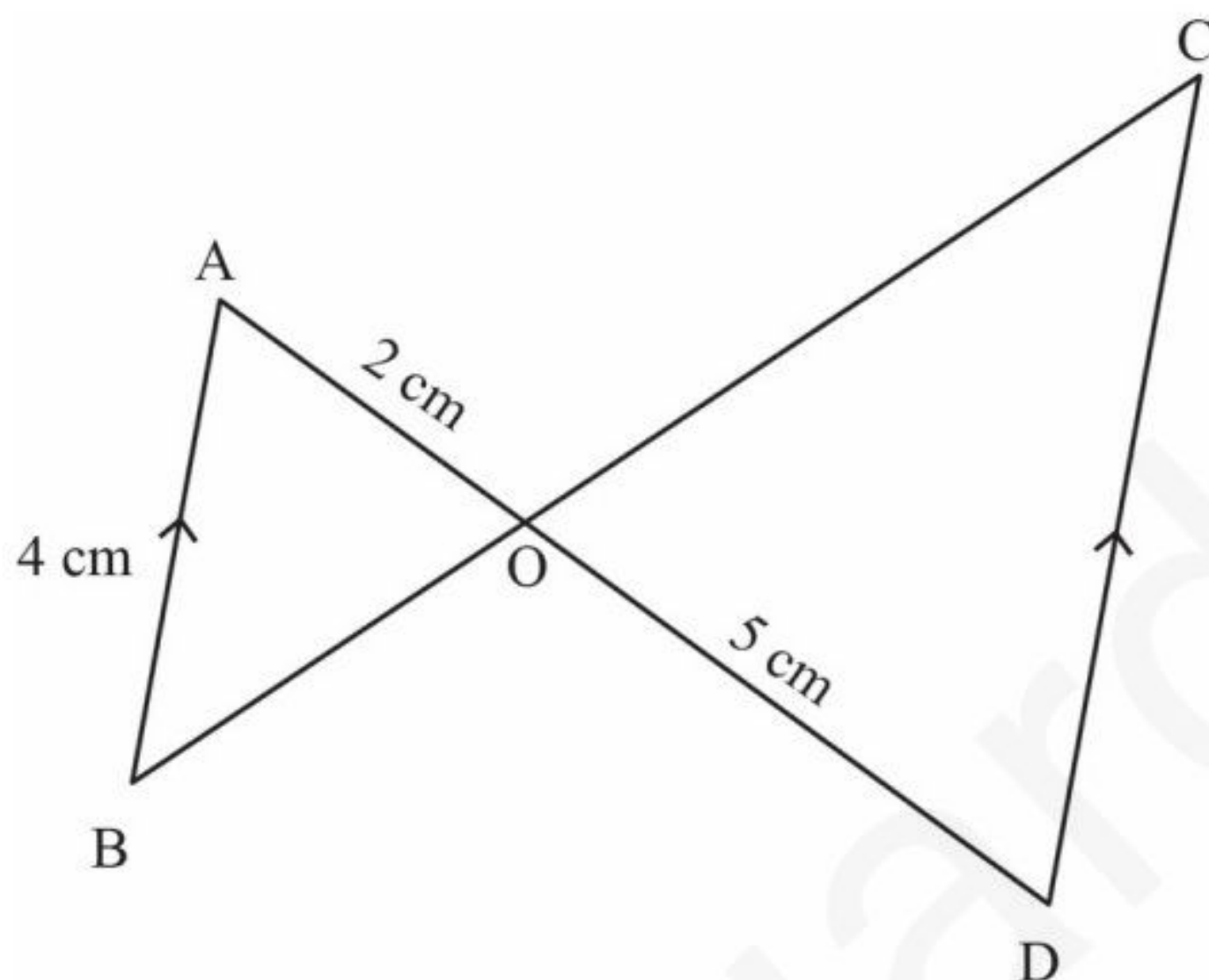
SECTION B: 48 MARKS]

Answer any **four** questions in this section.

Each question carries 12 marks.

- 6 (a) Find the Lowest Common Multiple (LCM) of $y^2 - 1$ and $y^2 - 3y - 4$. [3]

(b)



In the diagram AB and CD are parallel lines $OA = 2$ cm, $OD = 5$ cm and $AB = 4$ cm. The area of triangle AOB is 12 cm^2 .

- (i) Name the triangle similar to triangle AOB. [1]
- (ii) Calculate the area of triangle OCD. [2]
- (c) The masses, measured to the nearest gramme, of 15 randomly selected seeds are shown in the table below.

Mass (g)	5	6	7	8
No of seeds	4	3	6	2

- (i) State the modal mass. [1]
- (ii) Calculate the mean mass. [2]

- (d) Themba and Chipu take a driving test. The probability that Themba passes is $\frac{3}{4}$ and that Chipu passes is $\frac{2}{3}$.

Find the probability that

(i) Chipu fails, [1]

(ii) both fail, [2]

7 (a) (i) Express $\frac{1}{2} + \frac{3}{y+1}$ as a single fraction. [2]

(ii) Hence or otherwise solve the equation $\frac{1}{2} + \frac{3}{y+1} = 0$. [2]

- (b) Triangle DEF is such that $EF = 2x$ metres and the perpendicular distance of D from EF = $(6x - 10)$ metres. The area of triangle DEF is 16 m^2 .

(i) Form an equation in x and show that it reduces to $3x^2 - 5x - 8 = 0$. [2]

(ii) Solve the equation $3x^2 - 5x - 8 = 0$ using the method of completing the square. Give the answers to 2 decimal places. [5]

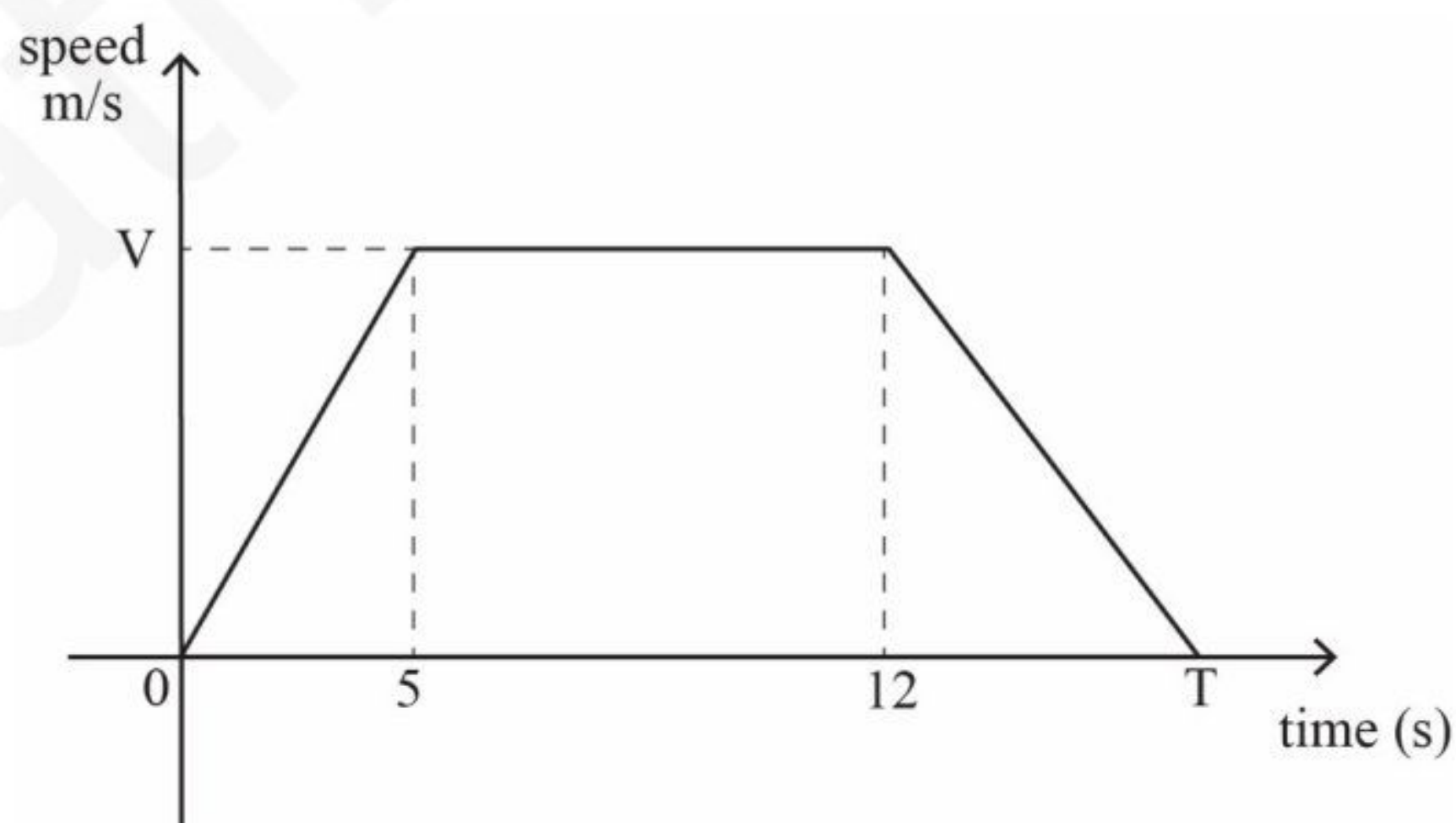
(iii) Hence find the perpendicular distance of D from EF. [1]

- 8 (a) The time Q , hours varies inversely as the speed P km/h.
When $Q = 24$ minutes, $P = 30$ km/h.

(i) Find the relationship between Q and P , [3]

(ii) Calculate the value of Q when $P = 3$ km/h. [1]

(b)



The graph represents the speed-time graph of an athlete during a training session. Acceleration during the interval from $t = 0$ to $t = 5$ is 3 m/s^2 .

- (i) Find the value of V . [2]
- (ii) Calculate the distance the athlete covered during the first 12 seconds. [2]
- (iii) Calculate T , given that the total distance covered is 165 m. [2]

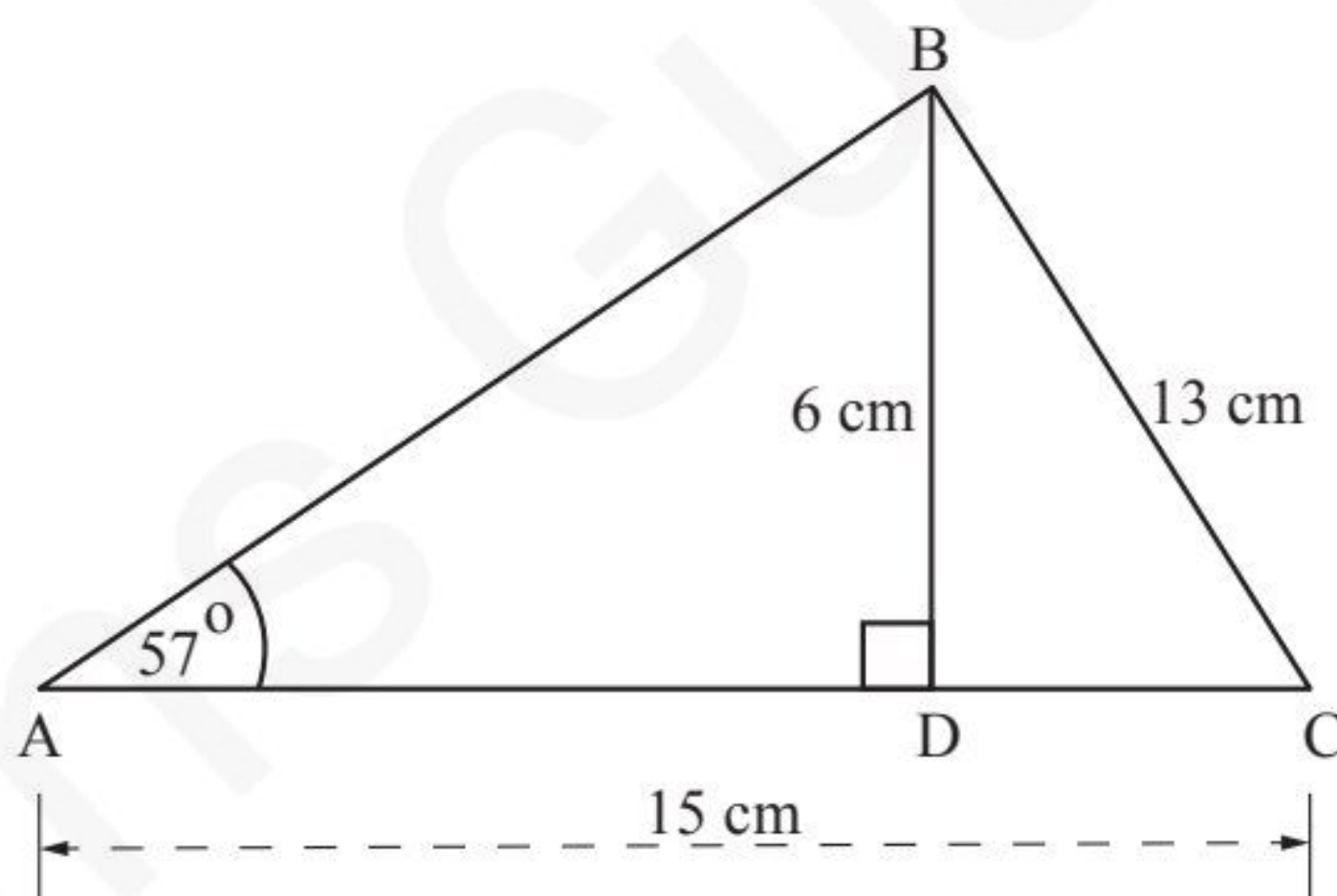
- (c) A solid cylinder has a radius of 10 cm and a height of 16 cm, calculate the total surface area of the cylinder.

[In this question take π to be 3.142] [2]

- 9 (a) The salary (S) of a gardener is partly constant and partly varies as the number of days (N) worked. \$60 is paid for working 25 days and \$40 is paid for working 15 days.

Calculate the salary after working for 10 days. [4]

- (b) ABC is a triangle with BD perpendicular to AC.



$AC = 15 \text{ cm}$, $BC = 13 \text{ cm}$, $BD = 6 \text{ cm}$ and $\hat{BAD} = 57^\circ$.

- (i) Calculate the length of DC. [2]
- (ii) Find $\hat{ABC}$. [4]
- (c) Matrix $A = \begin{pmatrix} -7 & -1 \\ -3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & -2 \\ 1 & -6 \end{pmatrix}$.

Find $A + B$. [2]

- 10 The table below shows the time (in minutes) spent by 100 randomly selected customers who entered into the bank.

Time (mins)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 50$	$50 < t \leq 60$
No of customers	5	30	p	25	12	5

(a) Find

(i) the value of p, [1]

(ii) an estimate of the mean time. [2]

Below is the cumulative frequency table for the distribution.

Time	≤ 10	≤ 20	≤ 30	≤ 40	≤ 50	≤ 60
Cumulative frequency	5	35	58	83	q	100

(b) Find the values of q. [1]

(c) Answer this part of the question on graph paper.

Use a scale of 2 cm to represent 10 minutes on the time axis and 2 cm to represent 10 customers on the cumulative frequency axes.

Draw a cumulative frequency curve for this distribution. [4]

(d) Use the graph to find the

(i) number of customers who spend 36 minutes in the bank, [2]

(ii) median time. [2]

- 11 Answer the whole part of this question on a sheet of graph paper.
Use a scale of 2 cm to represent 1 unit on both axes for $-1 \leq x \leq 7$ and $-1 \leq y \leq 6$. Triangle XYZ has vertices at X(1,2), Y(3,3) and Z(2,5).

(a) Draw and label triangle XYZ. [1]

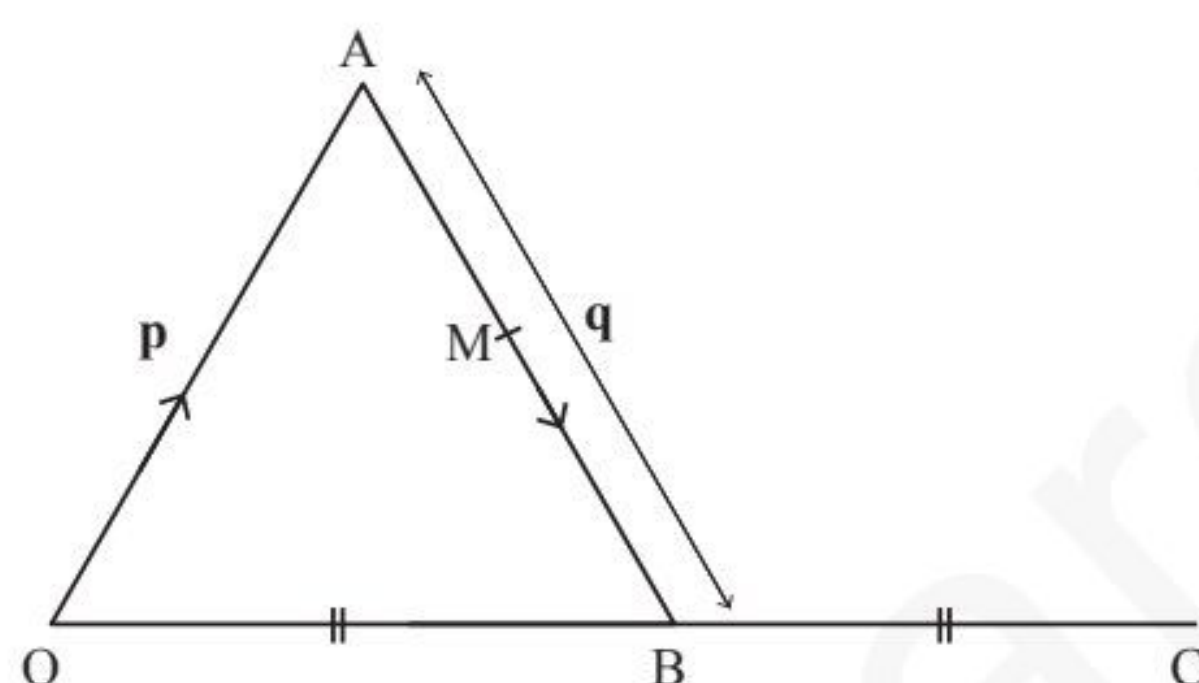
(b) $Z_1(5,4)$ is the image of Z under a translation T.

(i) Find the translation vector T. [1]

(ii) Draw and label triangle $X_1Y_1Z_1$ the image of triangle XYZ under T. [2]

- (c) Points $X_2(2,1)$ and $Y_2(3,3)$ and $Z_2(5,2)$ are the images of X , Y and Z under a transformation U
- (i) Draw and label triangle $X_2Y_2Z_2$ the image of triangle XYZ under transformation U . [2]
- (ii) Describe transformation U fully. [3]
- (d) Triangle $X_3Y_3Z_3$ is the image of triangle XYZ under a reflection in the y -axis.
Draw triangle $X_3Y_3Z_3$. [3]

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In the diagram, $\overrightarrow{OA} = p$, $\overrightarrow{AB} = q$, and M is the mid-point of AB . OB is produced to C such that $OB = BC$.

- (a) Express the following vectors in terms of p and/or q .
- (i) $\overrightarrow{OB}$, [1]
- (ii) $\overrightarrow{OC}$, [2]
- (iii) $\overrightarrow{OM}$, [1]
- (iv) $\overrightarrow{AC}$. [1]
- (b) OM is produced to a point T , not in the diagram, such that $OT = kOM$, where k is a constant.
Express $\overrightarrow{OT}$ in terms of k , p and q . [1]
- (c) If point T is on AC such that $AT = hAC$.
Find another expression for $\overrightarrow{OT}$ in terms of h , p and q . [1]
- (d) Using the results in (b) and (c), find the value of h and k . [3]
- (e) Find the ratio of $MT:OT$. [2]